

Effects of Integrating an AI Learning Companion into a Self-Regulated Learning 4L Framework on Elementary Mathematics Learning

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AI Learning Companions Require Regulatory Scaffolding



① AI and Intelligent Tutoring Systems in Education

Intelligent tutoring systems can support differentiated instruction through learner modeling, adaptive feedback, and learning analytics. (Chen et al., 2020; Labadze et al., 2023; Lin et al., 2023; Silva et al., 2022)

② GenAI as Digital Scaffold

Generative AI may serve as a digital scaffold for self-regulated learning by providing real-time feedback, prompts, and step-by-step guidance. (Chiu, 2024; Wu et al., 2023)

③ Potential Tension

Structured AI guidance may facilitate self-regulated learning, but excessive reliance on AI-generated responses may undermine learner agency. (Jin et al., 2023; Lee & Low, 2024)

④ Research Gap

It remains underexplored whether AI reliance and SRL engagement patterns differ across achievement levels. (Trinovita et al., 2025)

The Scaffold–Dependence Tension in GenAI-Supported Learning



Scaffold

- Real-time feedback
- Step-by-step guidance
- Planning, monitoring, and reflection
- Differentiated support

(Chiu, 2024; Wu et al., 2023)

Dependence

- Passive answer-seeking
- Procedural shortcutting
- Reduced learner agency

(Jin et al., 2023; Lee & Low, 2024)

Research gap:

It remains underexplored whether this scaffold–dependence tension manifests differently across achievement levels. This study addresses this gap by comparing TALPer-mediated regulatory engagement between high- and low-achieving learners.

Aligning SRL 4L with Zimmerman's SRL Cycle

Zimmerman (2002) SRL Cycle

Forethought

Goal Setting &
Prior-Knowledge Activation

Performance

Monitoring, Strategy Use &
Peer Adjustment

Self-Reflection

Error Noticing &
Conceptual Consolidation

Ho (2014) SRL 4L Framework

Self-learning

Goal Setting &
Prior-Knowledge Activation

Co-learning

Monitoring &
Scaffolded Feedback

Mutual learning

Cross-Group Strategy Sharing

Teacher-guided Learning

Misconception Correction &
Conceptual Integration

The SRL 4L phases map onto Zimmerman's cycle. This mapping informed the design of the SRL coding scheme and ENA dimensions adopted in the present study.

Taiwan Adaptive Learning Platform (TALP, developed by the MOE and NTCU since 2017)



TALP

Accounts

4M⁺

Students

220K⁺

Teachers

4235

Schools



Updated on June 12, 2026



adl.edu.tw

TALPer: The TALP AI Learning Companion (based on ChatGPT and Gemini)



T-TALPer

AI Assistant
for Teachers

TALPer was launched nationwide in
September 2024 and has reached
approximately 700,000 users.



Domain Specific
(S-TALPer)

AI Tutor
for Students

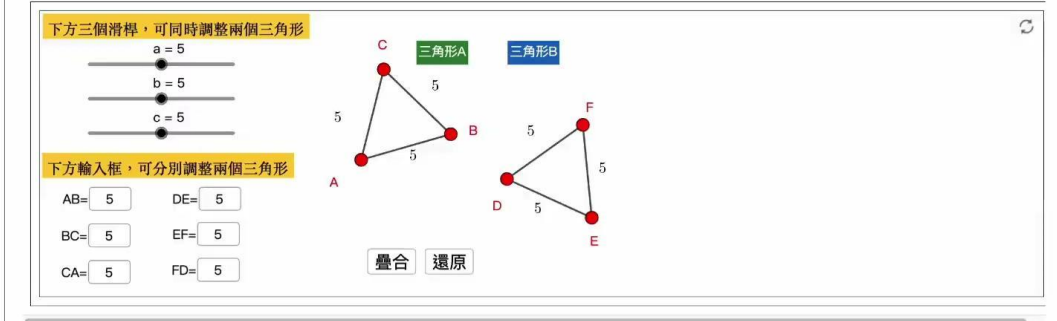
Domain General
(G-TALPer)

AI學伴互動討論區

e度: Hi! 小柚! 我是AI學伴e度, 讓我們一起來學習「三角形的全等性質與相似性質」這個單元吧! 請嘗試操作右邊畫面中間的兩個三角形, 試試看能否將兩個三角形完全疊合, 並將結果填列在作答區的表格中。

三角形的全等性質與相似性質

請嘗試操作下方的三角形, 試試看能否將兩個三角形完全疊合, 並將結果填列在作答區的表格中。



作答區

AB	BC	CA	DE	EF	FD	是否可以完全疊合 (請回答是或否)
5	6	5	5	5	6	
3	4	5	6	8	10	
4	7	9	4	7	9	
6	6	6	6	6	6	

送出

送出答案

Lin, T.-C., Shih, Y.-T., & Li, C.-H. (2026). Designing and Evaluating a 5E-Structured GenAI Coach for Guided Inquiry: A Pedagogy-to-Prompt Engineering Framework. *Education Sciences*, 16(3), 384. <https://doi.org/10.3390/educsci16030384>

Kuo, B. C., Bai, Z. E., & Lin, C. H. (2026). Developing an AI Learning Companion for Mathematics Problem Solving in Elementary Schools. *Computers & Education*, 240, 105463.

Research Questions



RQ1

Does TALPer-integrated SRL 4L improve learning outcomes compared with SRL 4L alone?

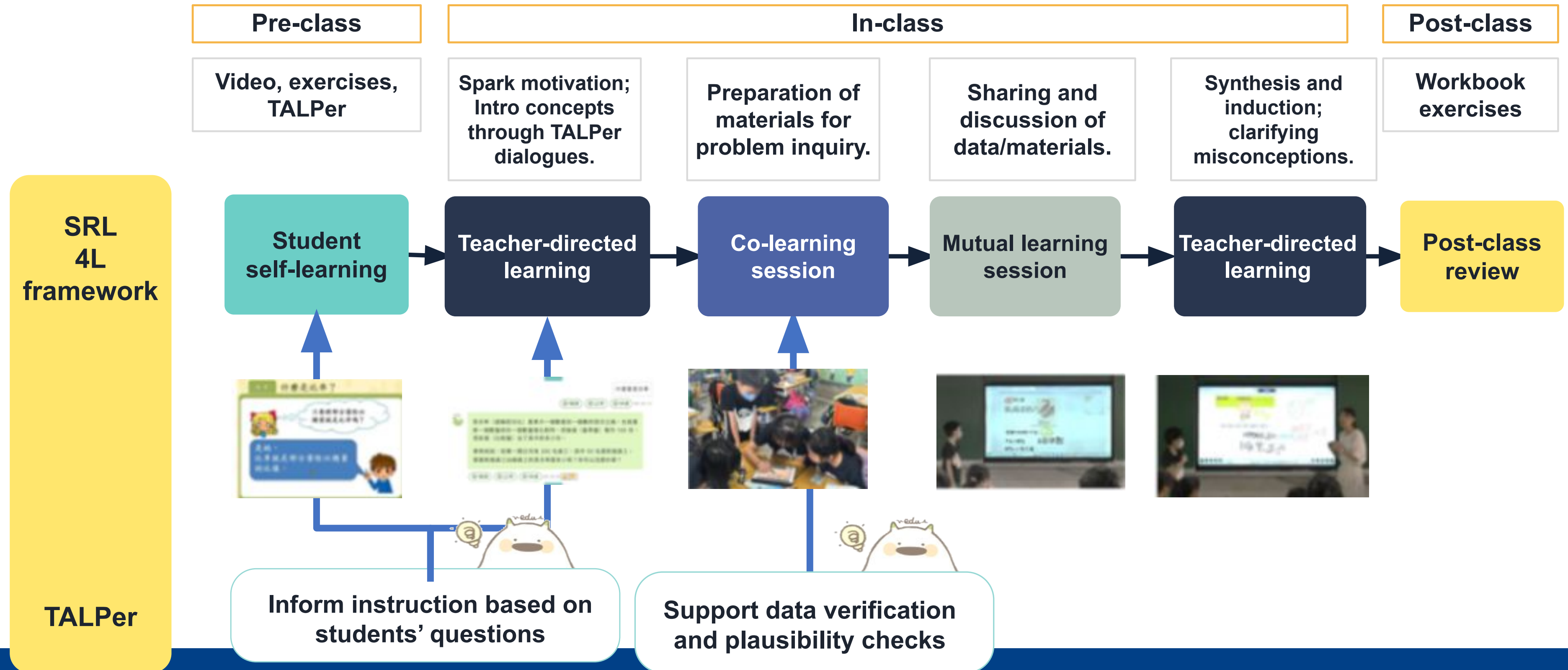
RQ2

Are there any differences in the application of SRL 4L between high- and low-achieving students?

RQ3

What differences in SRL cognitive network structures are revealed by ENA maps between high- and low-achieving students?

Pedagogical Framework: Integrating TALPer into SRL 4L



Research Design and Participants

44

fifth-grade students

Public elementary school, Taiwan

5

lessons on Ratios & Percentages

Delivered over 2 weeks

Comparison of the Two Groups

Experimental Group (n = 21)

- ▶ SRL 4L instruction with TALPer support throughout
- ▶ Pre-class: TALP videos and TALPer dialogues
- ▶ In-class: teachers leveraged TALPer logs
- ▶ TALPer was integrated across Co-, Mutual-, and Teacher-guided learning phases

Control Group (n = 23)

- ▶ Identical SRL 4L instructional structure
- ▶ Only difference: no TALPer access
- ▶ Pre-test scores: no significant group difference ($p = .376$)
- ▶ TALPer integration can thus be treated as the primary treatment condition.

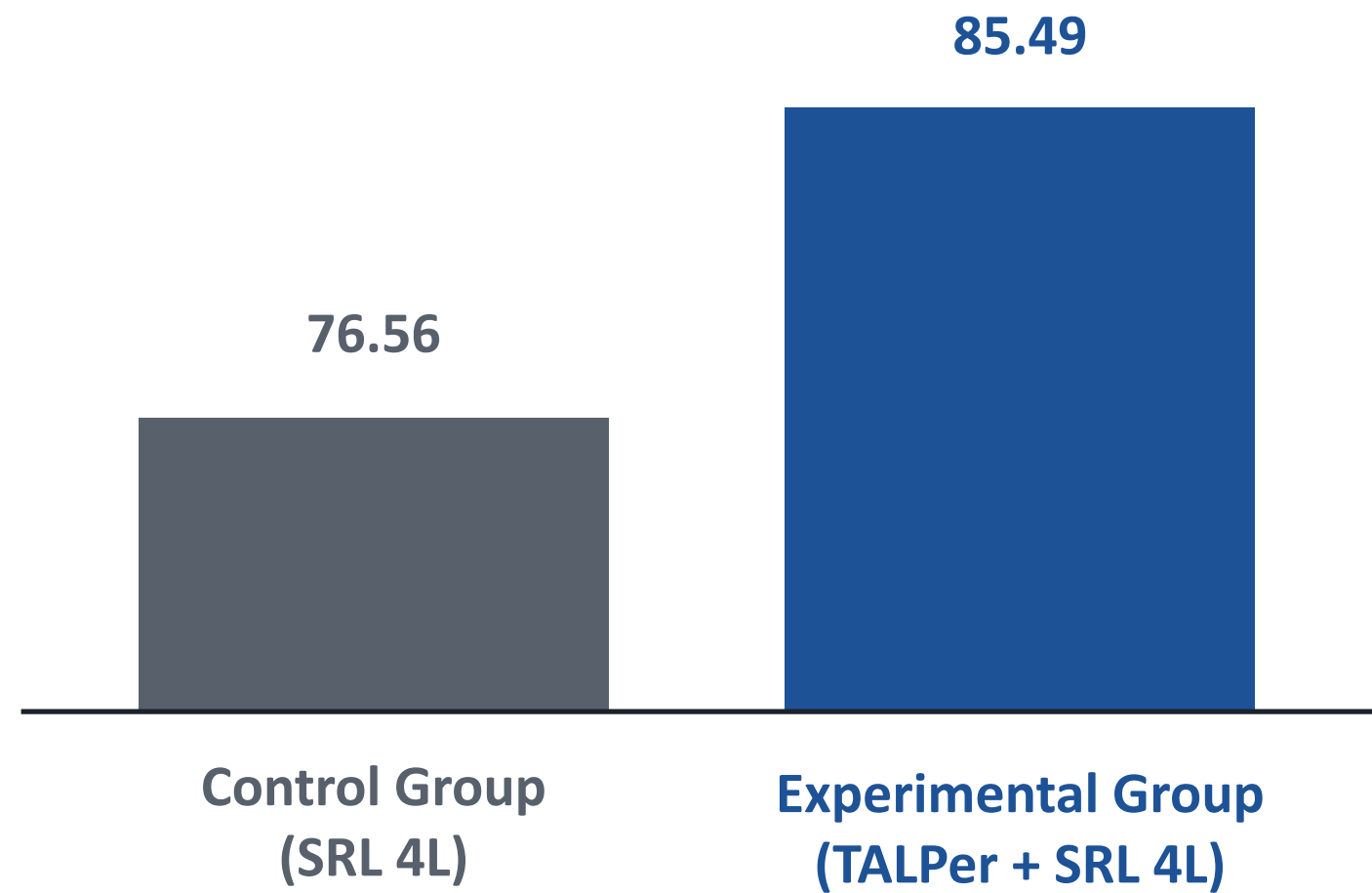
SRL Coding Framework and ENA Dimensions

Main category	Sub-category	Code	Explanation
Metacognition (M)	Setting Goals	M.SG	Proactively sets or adjusts learning objectives or sequences.
	Selecting Strategies	M.SS	Selects or modifies learning strategies, modalities, or task forms.
	Monitoring & Evaluation	M.ME	Real-time awareness and evaluation of conceptual understanding.
	Reflection	M.R	Reviews problem solving to identify errors or revise strategies.
Cognition (C)	Conceptual Understanding: Ask	C.CUA	Asks questions about concepts, principles, meanings, or applications.
	Conceptual Understanding: Explain	C.CUE	Articulates or constructs conceptual understanding.
	Procedural Knowledge: Ask	C.PKA	Asks about computational methods or problem solving steps.
	Procedural Knowledge: Execute	C.PKE	Performs actual computations or formulates equations/answers.

ENA maps code co-occurrences into a 2D space. In this model, the X-axis is interpreted as relative cognitive tendencies, whereas the Y-axis reflects differences in network complexity and

Results: TALPer-Supported SRL 4L Improved Post-Test Performance

Adjusted post-test means



ANCOVA result

$$F(1,41) = 4.71$$

$$p = .036$$

$$\eta^2 = .114$$

Pre-post gains: Control +14.45; TALPer +26.45 (both $p < .001$)

Results: TALPer Usage Patterns Differed Significantly by Achievement Level

High-achieving students

- Goal setting
- Conceptual questions
- Monitoring and reflection
- Iterative strategy use

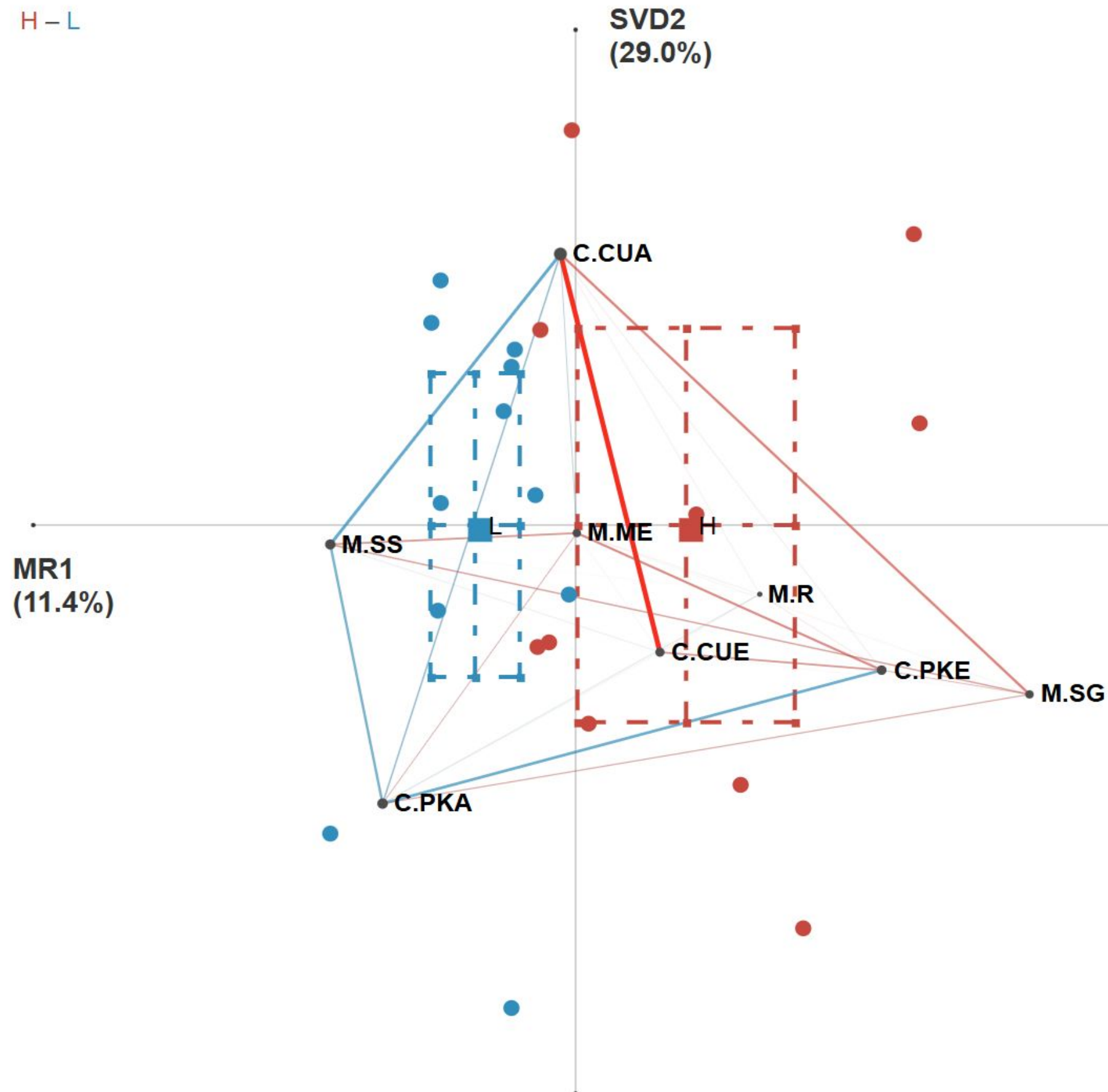
Low-achieving students

- Procedural strategy selection
- Procedural asking
- Answer-seeking without reflection
- Minimal self-monitoring

$t(12.07) = 4.05, p < .001, \text{Cohen's } d = 1.83$

Experimental group only: high-achievers (n = 10) vs. low-achievers (n = 11)
Comparison: ENA X-axis centroid difference between ability groups

Results: ENA Revealed Integrated vs. Fragmented Regulatory Networks



High-achieving students:
integrated, multidirectional
regulatory network

Low-achieving students:
fragmented, procedural,
ask-execute pattern

Conclusion & Implications



TALPer's effectiveness depends on learners' regulatory awareness.

- TALPer significantly improved learning outcomes within the SRL 4L framework.
- High-achievers used TALPer for conceptual inquiry, monitoring, and reflection.
- Low-achievers exhibited procedural dependence with limited reflective engagement.
- Teachers should scaffold students' strategic use of TALPer through guided questioning, structured reflection, and modeling productive AI-mediated dialogue.

Conclusion:

TALPer is most effective when learners engage it as a regulatory tool, rather than merely as a procedural shortcut.

Acknowledgments & Contact



Thank you. Questions and discussion welcome.

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